

Enable circular building design and deconstruction through an interoperable digital ecosystem for construction materials traceability.

Construction × Product traceability and digital product passport × Digital twin · OS055 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
31 is the world moving	10 can we play, can we win	MEDIUM 0 named in the evidence	€282m partial evidence · per year	LATER research and standards activi...	L3 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 1.0 published reference(s) in Construction, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

We can help construction firms and building owners meet EU circularity rules by linking Digital Product Passports (DPPs) with Building Information Modeling (BIM) to trace materials from design to deconstruction. This opportunity exists now because the EU's Ecodesign for Sustainable Products Regulation is pushing for DPPs, and research shows that integrating DPPs with BIM can enable circular construction and material reuse. For Orange Business, this is a chance to lead in a nascent market by combining our IoT expertise and research capabilities with digital twin technology.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€629m €32.9m – €3.1bn	€282m €14.7m – €1.4bn	€352k €18k – €1.7m	partial
Observed: tendered contract value, annualised	€123m €123m – €471m	€123m €123m – €471m	€154k €154k – €588k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Construction (EU27_2020)	250,045 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting Digital twin	2.9 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOTDPP	2021
Annual value of one engagement	€952k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	22,551 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E_IOTDPP is used as a proxy for Digital twin: IoT in production as the observable precursor. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

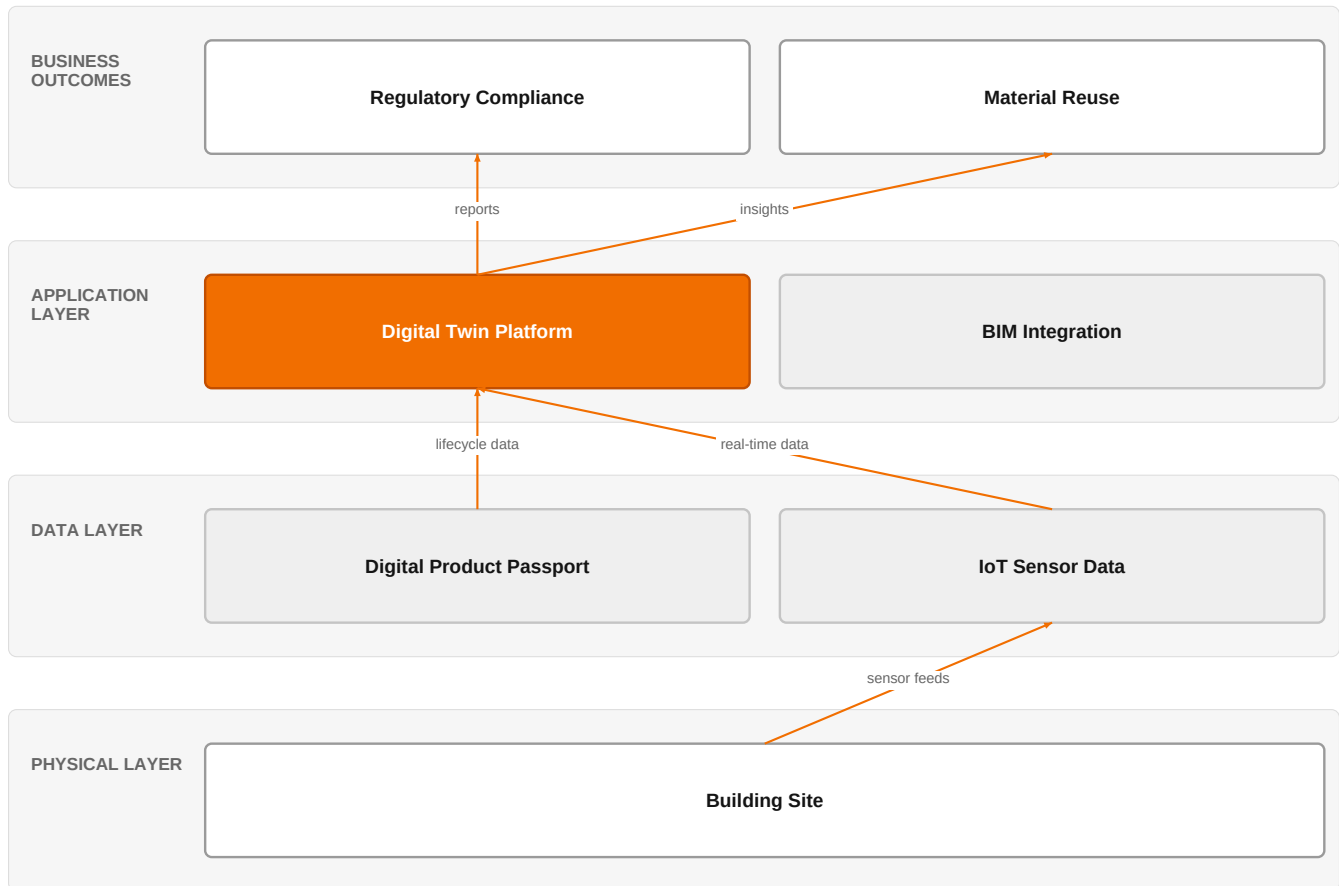
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€123m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 24 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Circular Building Traceability Ecosystem



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how Orange's digital twin platform connects IoT sensor data and digital product passports to BIM, enabling circular building practices.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU's Ecodesign for Sustainable Products Regulation (ESPR) mandates Digital Product Passports for electronics and appliances by 2027, creating an urgent need for machine-readable product lifecycle data to support the circular economy. This regulatory push is extending to construction, where buildings account for a significant share of energy consumption and CO2 emissions, and the industry struggles to implement sustainable practices despite available digital technologies. Research is now exploring how to integrate DPPs with BIM to enable circular construction and improve material traceability across building lifecycles, and projects like CRedIBIE are aiming to enable design, adaptability, reuse, and deconstruction of buildings through a fully interoperable digital ecosystem.

Sources: SIG-BA196FF05244 Springer Science and B 2026-02-03; SIG-955B81B681F1 cordis.europa.eu 2025-11-01; SIG-30E389C313E3 openalex.org 2025-12-01; SIG-9E793F703B1B cordis.europa.eu 2025-10-01

Who buys, and why now

The buyer is typically a sustainability or digital transformation lead at a large construction firm, a building owner, or a public authority responsible for infrastructure. The pain is operational: they need to track materials across complex supply chains and building lifecycles to comply with regulations and meet circularity goals, but they lack the data interoperability to do so. The trigger is often an upcoming regulatory deadline, a major renovation or deconstruction project, or a public tender that requires circularity credentials.

Why it is hot now — every claim bound to a dated source

- The CRedIBIE project aims to enable design, adaptability, reuse, and deconstruction of buildings through a fully interoperable digital ecosystem. [SIG-9E793F703B1B]
- CRedIBIE project enables design, adaptability, reuse, and deconstruction of buildings through a fully interoperable digital ecosystem. [SIG-9E793F703B1B]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution that combines its IoT experts and Orange Group researchers to design and deploy a digital twin platform for building material traceability. The engagement would start with a discovery phase to map the customer's existing BIM and supply chain data, then build a digital twin that integrates DPPs with BIM, enabling real-time tracking of materials from sourcing to deconstruction. Orange would also provide consulting on data interoperability and regulatory compliance, leveraging its research capabilities to stay ahead of evolving standards.

Why Orange can win

Orange can win because it brings a unique combination of IoT expertise, research capabilities, and a sovereign cloud offering (through its partnership with Dassault Systemes / 3DS Outscale) that addresses trust and sovereignty concerns. The assets are thin in terms of a ready-made construction-specific solution, but Orange's ability to integrate digital twin technology with IoT and its research depth give it a credible edge over pure IT consultancies.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Digital twin	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 5.9; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
Accenture	Global systems integrator	sells Digital twin; competes in OX: Smart Industries	structural	—
Capgemini	Global systems integrator	sells Digital twin; competes in OX: Smart Industries	structural	—
Dassault Systemes / 3DS Outscale	Sovereign cloud provider	sells Digital twin; competes in OX: Smart Industries	structural	—
PTC (ThingWorx)	Industrial / OT specialist	sells Digital twin; competes in OX: Smart Industries	structural	—
Schneider Electric	Industrial / OT specialist	sells Digital twin; active in Construction; competes in OX: Smart Industries	structural	—
Siemens	Industrial / OT specialist	sells Digital twin; active in Construction; competes in OX: Smart Industries	structural	—
AWS	Hyperscaler	competes in OX: Smart Industries — also an Orange partner	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 Are you currently tracking materials across your building projects for regulatory compliance or circularity goals?
- 2 Do you have a BIM model for your existing or planned buildings that could serve as a foundation for a digital twin?
- 3 What is your biggest pain point in material traceability: data silos, lack of interoperability, or regulatory pressure?

- 4 Have you started any pilot projects for digital product passports or digital twins in construction?
- 5 Who in your organization would own the budget for a digital traceability initiative: IT, sustainability, or operations?

Objections you will meet

They will say	You can answer
We already have a BIM system and don't see the need for a digital twin.	That's a fair point. BIM is a great foundation, but it's typically static. A digital twin adds real-time data from IoT sensors, enabling you to track materials throughout the building's life, which is essential for deconstruction and reuse.
The regulatory requirements for construction aren't as clear as for electronics.	You're right, the ESPR currently targets electronics, but the Construction Products Regulation is moving in the same direction. Getting ahead of this now will put you in a stronger position when mandates arrive.
This sounds expensive and complex to integrate with our existing systems.	It can be, but we start with a discovery phase to understand your current data landscape and identify quick wins. Our approach is incremental, and we leverage your existing BIM investment.

Risks and unknowns

The main risk is that the market is still nascent, with no established standards for DPP-BIM integration, and regulatory mandates for construction are not yet as concrete as for electronics. There is also uncertainty about customer readiness to invest in digital twins for traceability, given the complexity and cost. Orange's assets are limited, and it may need to partner to fill gaps in BIM expertise or construction domain knowledge.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Bid brief: develop a proposal outline for a construction industry consortium focused on circular building, leveraging Orange's digital twin L3 and IoT expertise, and include a section on how we would ensure interoperability with existing standards.
Sales	In a conversation with a European construction firm's sustainability director, say: 'We are exploring how digital twin technology can support circular building design and material traceability, and we'd like to understand your current challenges in tracking materials across a building's lifecycle.'
Strategist / Innovator	Deep-dive brief: map the CRedIBIE project's interoperability standards and data models to Orange's digital twin L3 capability, identifying gaps where Orange Group researchers and IoT experts could contribute to circular building traceability.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-AE9A7F9C36FF	2026-05-04	openalex.org	1	Computer Vision and Machine Learning Approaches for Defect Detection in 3D-Printed Cementitious...
SIG-F47C31AB60CB	2026-02-14	MDPI AG	1	Dynamic Traceability and Reliability Management: Integrating the Digital Product Passport with ...
SIG-BA196FF05244	2026-02-03	Springer Science and Busi...	1	Digital Product Passport for Circular Economy: A Review of Semantic Web Vocabularies and Requir...
SIG-30E389C313E3	2025-12-01	openalex.org	1	Integrating Digital Product Passports and BIM for Circular Construction: A Review of Challenges...
SIG-A598AE09F54F	2025-11-20	openalex.org	1	Real-time vision-based defect detection for large-scale on-site earthen additive manufacturing:...
SIG-955B81B681F1	2025-11-01	cordis.europa.eu	1	INTELLIGENT DIGITAL TOOLS FOR GREEN RENOVATION, ASSESSMENT, TRANSFORMATION, EFFICIENCY AND SUST...
SIG-EFEBD520406D	2025-10-15	openalex.org	1	Stakeholder perspectives on digital product passports for construction products
SIG-40DC35D6534F	2025-10-14	openalex.org	1	Predictive Generation of Digital Product Passport as Decision Support in Circular Networks
SIG-9E793F703B1B	2025-10-01	cordis.europa.eu	1	Circular REDesign for a Resource-efficient, Innovative, and Transformative Built Environment
SIG-6FBECA0F7EA7	2025-09-10	openalex.org	1	Design of a Sensor-Based Digital Product Passport for Low-Tech Manufacturing: Traceability and ...
SIG-6970406DD236	2025-06-01	cordis.europa.eu	1	Maintenance-oriented Digital Twin for Underground Infrastructure with Sensing, Machine Learning...
SIG-FF10DC5527D0	2025-06-01	cordis.europa.eu	1	Novel biobased formulations for CIRCULAR Construction materials
SIG-EB818F68D9F4	2024-11-01	cordis.europa.eu	1	Roll out Systemic Change to evolve the Furniture Industry towards CE through the Implementation...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:43:43+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.